

DATA SCIENCE & AI MASTERS 2026

120-Day Deep-Mastery Study Plan - 6 Hours/Day - 14 Phases

Plan element	Replanned target
Course basis	25 sections 632 lectures 12 quizzes 4 role-plays 115h 56m
Study commitment	6 focused hours per study day
Time available	120-day deep-learning window
Learning method	Learn -> reproduce -> independent practice -> review -> build -> explain
Primary outcome	Job-ready evidence: skills, projects, deployments, interview readiness

What changed from the previous PDF

The 120-day roadmap is now organized into exactly 14 phases, matching the structure of the earlier intensive plan while preserving the deeper 120-day pacing. The phases follow the source syllabus and separate each major skill domain so that Python, statistics, EDA, SQL, ML, time series, deep learning, NLP, GenAI, agents, deployment, capstones and career preparation each have a clear home.

Daily 6-hour framework

Block	Time	Purpose
A	90 min	New concepts / lectures
B	90 min	Code along + reproduce
Break	30 min	Reset
C	90 min	Independent coding / exercises
D	60 min	Notes + active recall + theory
E	30 min	Error log + quiz/review/planning

Rule: a lecture is input, not the work. Convert every important concept into code, notes, an independent task and an explanation.

14-Phase Master Roadmap

#	Phase	Source	Study days	Primary objective
1	Phase 1 - Programming Foundations	Section 02	Study Days 1-5	Python, NumPy, Pandas, visualization, OOP and independent coding
2	Phase 2 - Statistical Foundations	Section 03	Study Days 6-9	Descriptive/inferential statistics, probability, distributions, hypothesis testing
3	Phase 3 - Data Analysis & Exploration	Section 04	Study Days 10-12	EDA process, cleaning, scaling, outliers, feature transformation and reporting
4	Phase 4 - Database & Query Skills	Section 05	Study Days 13-15	SQL, joins, functions, subqueries, windows, stored procedures and Python connectivity
5	Phase 5 - Machine Learning	Section 06	Study Days 16-30	Regression, classification, ensembles, clustering, feature engineering and HPO
6	Phase 6 - Time Series Forecasting	Section 07	Study Days 31-33	Stationarity, ARIMA, Prophet, evaluation and independent forecasting
7	Phase 7 - Deep Learning & Neural Networks	Section 08	Study Days 34-40	ANN, CNN, RNN/LSTM, transfer learning and image projects
8	Phase 8 - Natural Language Processing	Section 09	Study Days 41-48	NLP preprocessing, embeddings, LSTM, GRU and bidirectional LSTM
9	Phase 9 - Transformers, LLMs & RAG	Section 10	Study Days 49-64	Attention, BERT/GPT, prompting, RAG, LangChain, vector DBs, Ollama and APIs
10	Phase 10 - Advanced Generative AI	Sections 11-12	Study Days 65-70	Fine-tuning, PEFT, LoRA/qLoRA and rigorous RAG evaluation
11	Phase 11 - Agentic AI & Automation	Sections 13-15	Study Days 71-79	Tools, tool calling, LangGraph, agents and n8n automation
12	Phase 12 - Production Engineering & Cloud	Sections 16-17, 23	Study Days 80-86	Flask, model serving, AWS EC2, ETL, data platforms and GenAI deployment
13	Phase 13 - Industry Capstone Projects	Sections 18-22	Study Days 87-101	ML, DL, GenAI and agentic portfolio projects with deployment and documentation
14	Phase 14 - Career Mastery & Final Consolidation	Sections 24-25 + Mastery	Study Days 102-120	Interviews, client role-play, portfolio, mock assessment and final mastery review

Mastery standard

By the end of a phase, the target is not merely video completion. For core topics, you should be able to recognize the concept, reproduce an implementation, apply it to a new problem and explain why it works, when to use it and its limitations.

Phase 1 - Programming Foundations

Source: Section 02 Study days: Study Days 1-5

Phase objective: Python, NumPy, Pandas, visualization, OOP and independent coding

Foundation checkpoint: independently load, clean, analyze and visualize a dataset with Python, Pandas and NumPy, and structure reusable logic using functions/classes.

Day	Focus	6-hour execution target
1	Python setup + fundamentals	Installation, Colab, variables, types, operators. Code every example.
2	Python collections	Lists, tuples, sets, dictionaries, indexing and slicing.
3	Control flow + functions	Loops, iterations, control structures, functions.
4	Reusable Python	Map/reduce/filter, file handling, OOP.
5	Python DS stack + mini-project	NumPy, Pandas, Matplotlib, Seaborn; build a small EDA script; quiz.

Phase 2 - Statistical Foundations

Source: Section 03 Study days: Study Days 6-9

Phase objective: Descriptive/inferential statistics, probability, distributions, hypothesis testing

Day	Focus	6-hour execution target
6	Statistics foundations	Types of data, descriptive vs inferential, sampling, population vs sample.
7	Central tendency + dispersion	Mean, median, mode, range, IQR, variance, SD, mean deviation.
8	Probability + distributions	Rules, conditional probability, Bayes, uniform/binomial/Poisson/normal, Z-scores.
9	Inference + review	Covariance, correlation, p-values, t/z/chi-square/ANOVA; problem set + quiz.

Phase deliverables

1-page statistics cheat sheet; solved problem set; Python implementations of core tests.

Phase 3 - Data Analysis & Exploration

Source: Section 04 Study days: Study Days 10-12

Phase objective: EDA process, cleaning, scaling, outliers, feature transformation and reporting

Day	Focus	6-hour execution target
10	EDA fundamentals	DA/DS process, sourcing, cleaning, missing values, scaling.
11	EDA transformation	Outliers, invalid data, univariate/bivariate/multivariate analysis, binning.
12	Independent EDA	Feature encoding, case study, EDA report; complete a fresh dataset end-to-end.

Phase 4 - Database & Query Skills

Source: Section 05 Study days: Study Days 13-15

Phase objective: SQL, joins, functions, subqueries, windows, stored procedures and Python connectivity

Day	Focus	6-hour execution target
13	SQL foundations	Installation, architecture, constraints, DDL, DQL, DML, joins.
14	SQL advanced	Import/export, aggregation, string/date functions, regex, nested queries, views.
15	SQL mastery	Stored procedures, window functions, SQL-Python connectivity; 15+ queries + quiz.

Phase 5 - Machine Learning

Source: Section 06 Study days: Study Days 16-30

Phase objective: Regression, classification, ensembles, clustering, feature engineering and HPO

Largest technical phase. Protect time for independent model comparison, feature engineering, hyperparameter optimization and error analysis.

Day	Focus	6-hour execution target
16	ML fundamentals	Types of ML, use cases, features, train-test split, scaling, encoding.
17	Regression metrics	MAE/MSE/RMSE-style evaluation, residual thinking, regression workflow.
18	Linear regression	Simple + multiple linear regression; practical implementation.
19	Nonlinear + regularization	Polynomial regression, bias-variance, Ridge, Lasso.
20	Classification foundations	Log loss, confusion matrix, AUC-ROC, classification report.
21	kNN + decision trees	Theory, distance intuition, entropy, Gini, practicals, visualization.
22	Tree ensembles	Random Forest, Naive Bayes, SVM, Logistic Regression.
23	Classification project	Build and evaluate a complete classification solution.
24	Ensemble learning	Bagging, boosting, AdaBoost, Gradient Boosting, XGBoost.
25	Clustering	k-Means, hierarchical clustering, Mean Shift; interpret clusters.
26	Feature engineering I	RFE, SFS, Chi-Square and practical feature selection.
27	Feature engineering II	PCA, LDA, kPCA and QDA; interpretation and practicals.
28	Hyperparameter optimization	Manual HPO, GridSearch, RandomizedSearch, validation design.
29	Independent ML project	One dataset: pipeline, 3 model families, HPO, final model choice.
30	ML mastery checkpoint	Closed-book review, code from memory, quiz, gap-fixing.

Phase deliverables

1 complete ML project; model comparison table; feature engineering notes; tuned final model.

Phase 6 - Time Series Forecasting

Source: Section 07 Study days: Study Days 31-33

Phase objective: Stationarity, ARIMA, Prophet, evaluation and independent forecasting

Day	Focus	6-hour execution target
31	Time-series foundations	Components, decomposition, anomaly detection, stationarity and transformations.
32	ARIMA mastery	AR/MA, ACF/PACF, p/d/q, practicals, grid search, final model.
33	Prophet + forecasting project	Prophet, multivariate TS, MSE/RMSE/MAPE; independent forecast comparison.

Phase 7 - Deep Learning & Neural Networks

Source: Section 08 Study days: Study Days 34-40

Phase objective: ANN, CNN, RNN/LSTM, transfer learning and image projects

Day	Focus	6-hour execution target
34	Deep learning foundations	Neuron, activation functions, forward pass, backprop, gradient descent.
35	ANN practicals	ANN implementation, optimization, debugging and intuition.
36	CNN fundamentals	Convolution, pooling, architecture, augmentation, batch/epochs.
37	CNN practicals	Build CNN, inspect parameters, evaluate model, image pipeline.
38	RNN + LSTM	RNN types, vanishing/exploding gradients, LSTM gates and practicals.
39	Transfer learning	VGG16, MobileNet, transfer learning workflow.
40	Deep learning project	Retrain an image model on a new dataset; quiz + review.

Phase 8 - Natural Language Processing

Source: Section 09 Study days: Study Days 41-48

Phase objective: NLP preprocessing, embeddings, LSTM, GRU and bidirectional LSTM

Day	Focus	6-hour execution target
41	NLP foundations	Intro, linguistics, case folding, cleanup, tokenization, stop words.
42	Classical NLP	n-grams, vectorization, Bag of Words, TF-IDF.
43	Information extraction	POS tagging and NER; practical pipeline.
44	Word2Vec	CBOW, Skip-Gram, pretrained Word2Vec, embeddings intuition.
45	GloVe + FastText	Compare embedding families; cosine similarity practicals.
46	LSTM for NLP	Architecture, pointwise operations, forget/input/output gates.
47	GRU + BiLSTM	GRU gates, practicals, bidirectional LSTM.
48	NLP independent practice	Build a text classification/similarity task; quiz + review.

Phase 9 - Transformers, LLMs & RAG

Source: Section 10 Study days: Study Days 49-64

Phase objective: Attention, BERT/GPT, prompting, RAG, LangChain, vector DBs, Ollama and APIs

GenAI checkpoint: understand the architecture, build without the tutorial, then evaluate the system rather than stopping at a demo.

Day	Focus	6-hour execution target
49	Transformer architecture	Transformer types, architecture, self-attention.
50	Encoder/decoder understanding	Encoder, contextual embeddings, decoder architecture.
51	BERT family	BERT configs, fine-tuning, masked LM, RoBERTa, DistilBERT, ALBERT.
52	GPT architecture	Decoder-only GPT, masked multi-head attention, blocks, training.
53	LLM foundations	Tokens, context window, prompts, prompt engineering, prompt structures.
54	RAG fundamentals	What/why, use cases, retrieval architecture and generation flow.
55	RAG practical I	Build PDF RAG chatbot; understand ingestion, chunking and retrieval.
56	LangChain + prompting	LangChain, few-shot, chain-of-thought concepts and prompt design.
57	Vector databases	Vector DB vs index, similarity search, Pinecone practicals.
58	LSH + retrieval quality	LSH, retrieval experiments, error analysis.
59	Ollama	Setup, model testing, Python implementation.
60	Ollama RAG	Build and debug local RAG system.
61	LLM APIs	API patterns, structured prompting, integration.
62	xAI/Grok RAG	Implement API-backed RAG workflow.
63	Independent RAG build	Build from an empty folder without the tutorial.
64	GenAI mastery checkpoint	Quiz, architecture review, failure modes, retrieval debugging.

Phase deliverables

Independent PDF RAG chatbot; architecture diagram; retrieval error log.

Phase 10 - Advanced Generative AI

Source: Sections 11-12 Study days: Study Days 65-70

Phase objective: Fine-tuning, PEFT, LoRA/qLoRA and rigorous RAG evaluation

GenAI checkpoint: understand the architecture, build without the tutorial, then evaluate the system rather than stopping at a demo.

Day	Focus	6-hour execution target
65	Fine-tuning vs RAG	Decision framework and use cases.
66	PEFT + quantization	PEFT, quantization, LoRA concepts.
67	LoRA + qLoRA	Understand adapters, low-rank updates and qLoRA workflow.
68	RAG evaluation metrics I	BLEU, ROUGE, METEOR, perplexity.
69	RAG evaluation metrics II	BERTScore, BARTScore, RAGAS.
70	Evaluation project	Score your own RAG system and write an evaluation report; quiz.

Phase 11 - Agentic AI & Automation

Source: Sections 13-15 Study days: Study Days 71-79

Phase objective: Tools, tool calling, LangGraph, agents and n8n automation

Day	Focus	6-hour execution target
71	Agentic AI foundations	Agents, examples, tools and workflow patterns.
72	Custom tools	Tool Decorator, StructuredTool, BaseTool, toolkits.
73	Tool calling + execution	Tool calling, execution loops, failure handling.
74	Agent mini-project	Currency conversion tool + travel planning agent.
75	Agent practicals	Code walkthrough, reasoning/planning/autonomy patterns.
76	LangGraph foundations	Stateful workflows, autonomy, planning, reasoning, adaptability.
77	LangGraph mini-project	Build and debug an agentic workflow.
78	n8n foundations	Templates, nodes, workflow design, tool integrations.
79	n8n AI automation	Build AI-powered news summarizer; connect an agent tool.

Phase 12 - Production Engineering & Cloud

Source: Sections 16-17, 23 Study days: Study Days 80-86

Phase objective: Flask, model serving, AWS EC2, ETL, data platforms and GenAI deployment

Day	Focus	6-hour execution target
80	Deployment fundamentals	Deployment architecture, Flask basics.
81	ML API	Build Flask model-serving application.
82	AWS fundamentals	EC2, networking concepts, deployment sequence.
83	AWS deployment	Deploy a model independently to EC2.
84	Data engineering basics	ETL, data warehouse, data marts, data lake; quiz.
85	GenAI deployment	Model serving, directory/PDF based systems, Streamlit.
86	GenAI on AWS	Deploy chatbot/application to EC2; document the architecture.

Phase 13 - Industry Capstone Projects

Source: Sections 18-22 Study days: Study Days 87-101

Phase objective: ML, DL, GenAI and agentic portfolio projects with deployment and documentation

Portfolio rule: build for understanding first, then polish 2-4 flagship projects with README, architecture, evaluation and deployment evidence.

Day	Focus	6-hour execution target
87	Cancer Risk - discovery	Problem statement, architecture, data understanding, EDA.
88	Cancer Risk - modeling	Model building, comparisons, XGBoost.
89	Cancer Risk - optimization	Optuna, model saving, validation and error analysis.
90	Cancer Risk - deployment	Streamlit + AWS + README.
91	Solar defect - baseline	Data understanding + base CNN.
92	Solar defect - transfer learning	MobileNetV2 + EfficientNetB0 + HPO.
93	Solar defect - deployment	Streamlit + EC2 + project documentation.
94	Elephant classification - baseline	Data + base model.
95	Elephant classification - transfer learning	MobileNetV2 + Xception.
96	Elephant classification - deployment	Streamlit + AWS + comparison report.
97	GenAI projects 1-2	ChatScholar + Research RAG chatbot; understand architecture patterns.
98	GenAI projects 3-4	Claims processing + Multi-PDF RAG.
99	GenAI projects 5-6	AI Career Coach + Sustainability Chatbot.
100	GenAI projects 7-8	Text-to-SQL + AWS Bedrock RAG.
101	GenAI project 9 + agentic	Resume Genie + Dental Appointment System; select flagship projects.

Phase deliverables

Selected flagship projects with GitHub-ready README, deployment notes and business explanation.

Phase 14 - Career Mastery & Final Consolidation

Source: Sections 24-25 + Mastery Study days: Study Days 102-120

Phase objective: Interviews, client role-play, portfolio, mock assessment and final mastery review

Career rule: answer from understanding, not memorized scripts. Record yourself, run mock interviews and practice client discovery + solution explanation.

Day	Focus	6-hour execution target
102	ML interviews	Study and explain 10 ML questions; build your own answers.
103	ML mock interview	Closed-book technical interview + review.
104	DL interviews	Study 10 DL questions and explain architectures.
105	DL mock interview	Closed-book interview + diagram explanations.
106	GenAI interviews I	Transformers, LLM, prompting, embeddings, RAG.
107	GenAI interviews II	Fine-tuning, evaluation, agents, deployment.
108	Full technical mock	ML + DL + GenAI under interview conditions.
109	Client role-play	RAG chatbot, churn model, website chatbot, demand forecasting.
110	Python-Statistics review	Rebuild a data workflow without notes.
111	EDA-SQL review	Complete data cleaning + SQL analysis task.
112	ML review	Build a compact end-to-end ML pipeline from scratch.
113	DL review	Rebuild an image model and explain choices.
114	NLP review	Build a small text task and compare representations.
115	RAG review	Build RAG from an empty project folder.
116	RAG evaluation review	Measure retrieval/generation quality; interpret failures.
117	Agentic review	Build a tool-calling agent + automation.
118	Deployment review	Deploy one project and verify it end-to-end.
119	Master assessment	Six-hour no-tutorial technical challenge: build, test, explain, document.
120	Final mastery + portfolio	Capability map, portfolio selection, resume/LinkedIn update, next-role plan.

Phase deliverables

Recorded mock answers; client role-play notes; final portfolio + resume/LinkedIn update.

Final 120-Day Mastery Checklist

Area	End-state target
Python / NumPy / Pandas	Independent implementation
Statistics	Strong conceptual + practical
EDA / SQL	Independent analysis workflow
Machine Learning	Strong model selection, evaluation and HPO
Time Series	Working forecasting capability
Deep Learning	Working ANN/CNN/LSTM/transfer-learning capability
NLP	Working classical + sequence modeling capability
Transformers / LLMs	Strong conceptual + practical understanding
RAG / Evaluation	Strong practical build + evaluation
Agentic AI / LangGraph / n8n	Working tool-calling and automation capability
Deployment / AWS	Working deployed application
Portfolio	2-4 flagship, documented projects
Interview / consulting	Ready for technical + client conversations

Completion standard

Do not measure success only by the percentage of lectures watched. Measure success by what you can build without the instructor, what you can explain without notes, what you can debug when it fails, and what you can show an employer as evidence.